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**SELF-SERVICE PRINTING STATION IN SKSU – ISULAN LIBRARY WITH
CASHLESS PAYMENT INTEGRATION VIA GCASH**

**NATHANIEL C. LAGNAODA
GEORGE ANDRIE D. GATAN**

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TRANSMITTAL

The thesis attached hereto entitled “**SELF-SERVICE PRINTING KIOSK IN SKSU – ISULAN LIBRARY WITH CASHLESS PAYMENT INTERGRATION VIA GCASH**” prepared and submitted by **NATHANIEL C. LAGNAODA** and **GEORGE ANDRIE D. GATAN**, in partial fulfillment of the requirements for the degree *Bachelor of Science in Computer Engineering* is hereby endorsed for approval.

JADE A. BUENAVIDES

Adviser

Date Signed

LENMAR T. CATAJAY, ME-CoE

College Dean

Date Signed

ELMER C. BUENAVIDES, DIT

Campus Research Coordinator

Date Signed

Accepted as partial fulfillment of the requirements for the degree of *Bachelor of Science in Computer Engineering*.

ROMMEL M. LAGUMEN

Campus Director

Date Signed

APPROVAL SHEET

The thesis attached hereto entitled “**SELF-SERVICE PRINTING KIOSK IN SKSU – ISULAN LIBRARY WITH CASHLESS PAYMENT INTERGRATION VIA GCASH**” prepared and submitted by **NATHANIEL C. LAGNAODA** and **GEORGE ANDRIE D. GATAN**, in partial fulfillment of the requirements for the degree *Bachelor of Science in Computer Engineering* is hereby endorsed for approval.

BEN ARNEL
Member

Date Signed

MEILAFLORE A. PACLIBAR
Member

Date Signed

JADE A. BUENAVIDES
Adviser

Date Signed

Approved as partial fulfillment of the requirements for the degree of *Bachelor of Science in Computer Engineering*.

LENMAR T. CATAJAY, ME-CoE
College Dean

Date Signed

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

Date Signed

ROMMEL M. LAGUMEN
Campus Director

Date Signed

BIOGRAPHICAL DATA



The researcher was born on September 22, 2003, in General Santos City he is youngest child of Mr. George Y. Gatan and Mrs. Marry Ann D. Gatan. He is fondly called “Adi,” “Gats,” and “Drei” by his family and friends.

He spent his childhood in his birthplace and completed his elementary education at Montessori Casa Precious Jewels Inc, Sto Niño, South Cotabato, in 2016. He persued his junior high school at Montessori Casa Precious Jewels inc and graduated in 2019-2020. For his senior high school, he studied at Sto Niño National High School under the Science, Technology, Engineering, Mathematics (Stem), which he completed in 2022. At present, he is pursuing a Bachelor of Science in Computer Engineering at Sultan Kudarat State University, Isulan Campus, Where he continues to enhance his technical and analytical skills

Looking ahead, the researcher aims to continue improving himself as he faces new opportunities and challenges with determination, hard work, faith in God, he hopes to achieve in his chosen career in the field of technology, possibly focusing in software development, systems design, or automation.

BIOGRAPHICAL DATA



The researcher, Nathaniel C. Lagnaoda, was born on December 12, 2003, at Quijano Clinic and Hospital in Tacurong City. He is the son of Mr. Bonifacio T. Lagnaoda Jr. and Mrs. Belinda C. Lagnaoda and is fondly called “Nikol” by his family and friends. He grew up with a curious mind and a strong passion for technology, showing early interest in computers and design. His upbringing instilled in him the values of hard work, perseverance, and humility, which continue to guide him in both his academic and personal life.

He began his educational journey at Masiag Central Elementary School, where he completed his elementary education. He then pursued his junior and senior high school studies at Notre Dame of Masiag, Inc., where he further honed his academic foundation and leadership skills. Currently, he is a fourth-year student taking up the Bachelor of Science in Computer Engineering at Sultan Kudarat State University, Isulan Campus. Throughout his college years, he has developed skills in programming, AutoCAD, circuit simulation, and basic hardware design using Arduino and ESP32 microcontrollers. He is also an active student

leader, serving as the Business Manager of the Engineering Student Organization, Class Representative of ICpEP.SE, and President of the Book Club.

Looking ahead, Nathaniel aspires to become a skilled and innovative computer engineer who can contribute to the advancement of technology and make a meaningful impact in his community. He plans to specialize in fields such as network engineering, automation, or embedded systems development. With dedication, continuous learning, and faith in God, he envisions building a successful career in the tech industry and possibly establishing his own technology-based enterprise to help empower others through innovation and education.

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Researchers

NATHANIEL C. LAGNAODA

GEORGE ANDREI D. GATAN

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ABSTRACT

NATHANIEL C. LAGNAODA and GEORGE ANDRIE D. GATAN, Month And Year Published. “SELF-SERVICE PRINTING KIOSK IN SKSU-ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH”. A thesis of Bachelor of Science in Computer Engineering, Sultan Kudarat State University, Isulan Campus, Isulan, Sultan Kudarat.

ADVISER: JADE A. BUENAVIDES

In this study, the researchers developed and implemented a Self-Service Printing Kiosk integrated with cashless payment via GCash, designed to provide students with a faster, more convenient, and efficient printing experience within the SKSU–Isulan Library. The kiosk automates the document printing process by allowing users to upload their files, convert them to PDF, preview the document, and pay electronically through GCash before printing. The system utilizes a Raspberry Pi microcontroller, touchscreen monitor, and thermal printer, all integrated through a custom software interface for smooth operation.

The project aims to address the challenges of long queues, manual payments, and the lack of accessible printing facilities within the library. Test results and observations show that the system successfully streamlines the printing process, minimizes human intervention, and promotes a secure, contactless transaction environment. This innovation enhances operational efficiency in the university library and aligns with modern digital payment trends, providing a sustainable and user-friendly solution for students and staff alike.

Chapter I

INTRODUCTION

Background of the Study

Educational institutions today face increasing demand for printing and document-handling services from students, faculty, and staff. Traditional manual print stations often lead to delays, long queues, and inefficient workflow, especially during peak periods. In the context of the Sultan Kudarat State University – Isulan Campus (SKSU-Isulan) library, the absence of an automated, cashless, self-service printing kiosk has resulted in bottlenecks, extended wait times, and higher dependence on library staff to manage printing and payment transactions. These issues highlight the need for a more efficient and user-friendly solution to accommodate the academic community's printing needs.

Kim, Yang, & Lee's (2023) study has shown the effectiveness of self-service kiosk systems in improving process efficiency and user satisfaction. For example, in the Republic of Korea, researchers found that the adoption of self-service kiosks in restaurants helped organizations become more resilient during the COVID-19 pandemic by reducing contact, supporting cashless transactions, and speeding service delivery. Another international study by Chung & Park (2021) evaluated the usability of self-service kiosks by older adults and found that design features such as side/back partitions and seating significantly reduced user workload and improved performance in kiosk use. These international results support the notion

that self-service kiosk introduction can enhance service delivery, improve user experience, and streamline operations.

Locally in the Philippines, mobile and kiosk-based innovations are likewise gaining attraction. A study on customer-perceived satisfaction in self-service kiosks in fast-food chains found significant positive effects of ease of use, navigability, and time-saving features on user satisfaction (Jherald et al., 2025). Additionally, a report on Office (2025) the first self-service print kiosk in the Philippines highlights how print-vending kiosks offer convenience, privacy, and easy payment to users, while generating new revenue streams and reducing labor costs for operators. These local insights underline the potential of self-service printing kiosks in the Philippine context, bridging user needs and technological opportunities.

Focusing on the Mindanao region as the base for this study, there is increasing institutional interest in self-service systems. For example, at Bukidnon State University (BukSU) (2023), a self-check kiosk for library transactions was recently introduced to improve operational efficiency and user convenience. Also, at the University of Southern Mindanao (USM), the study by Tutor (2025) found that the print-on-demand system was benchmarked by other institutions in Davao del Sur as a cost-effective and flexible digital printing service for academic use. These developments indicate a readiness and institutional appetite in Mindanao for print-service innovations, but they still fall short of fully automated, cashless, self-service printing kiosks integrated with mobile wallet payments.

Moreover, the effectiveness of a self-service printing kiosk with a cashless payment system lies in its potential to reduce manual handling of transactions, shorten wait times for printing, improve user satisfaction through convenience, and provide library administrators with usage reports for better monitoring. By enabling users to independently upload files, pay via mobile wallet (such as GCash), and receive printed output with minimal staff intervention, such a system aligns with modern academic service delivery and digital transformation objectives.

However, the gap remains in the limited adoption of such technology within Sultan Kudarat State University (SKSU). Despite SKSU's continuous efforts in technological advancement, the university library still lacks a self-service printing kiosk integrated with a secure, cashless payment system.

Therefore, this study aims to design, develop, implement and evaluate a self-service printing kiosk station in the SKSU-Isulan Library that integrates a cashless payment system via GCash.

Objectives of the Study

General Objectives

This study generally aims to develop a self-service printing station in SKSU-Isulan library that provides students and users with a convenient, automated, and cashless payment service through GCash.

Specific Objectives

1. Design a web-based system for document handling and printing.

2. Allow platform for seamless and cashless transaction through Gcash.
3. Generate data analytics for sales, transaction logs, usage and printing volume.
4. Evaluate the system in terms of:
 - 4.1. Functionality
 - 4.2 Reliability
 - 4.3 User Convenience

Significance of the Study

This study is designed to develop and implement a Self-Service Printing Kiosk in the Sultan Kudarat State University – Isulan Campus Library, integrated with a cashless payment system through GCash. The primary goal is to provide students, faculty, and staff with a convenient, automated, and secure method of printing documents without the need for manual transactions or cash handling. The system contributes to promoting digital transformation and technological innovation in campus services. Specifically, this study is significant to the following:

Students. This system offers convenience and efficiency to students by allowing them to print documents anytime without relying on library staff. Through cashless payment via GCash, students can transact quickly and securely, saving both time and effort. The kiosk helps address issues of long queues and manual payment processes, enhancing their overall learning and library experience.

Library Staff and Administrators. The kiosk automates printing and payment operations, minimizing manual supervision and reducing workload. It also provides automated transaction logs, weekly and monthly reports, and usage summaries that can be used to monitor revenue, system utilization, and performance. This enables better management and transparency in library services.

University Management. The project demonstrates the institution's commitment to digital innovation and modernization. By adopting an automated and cashless system, the university promotes environmental sustainability through paperless payment and resource efficiency. It also enhances SKSU's reputation as a forward-thinking university that embraces financial technology and automation for public service improvement.

Researchers. This study is significant to the researchers as it enhances their technical and analytical skills in the field of computer engineering. Through the design and implementation process, the researchers gain hands-on experience in system integration, hardware and software interfacing, and payment gateway development. It also strengthens their knowledge of user-centered design, cybersecurity, and system evaluation.

Future Researchers. This study may serve as a useful reference for future researchers who wish to explore similar innovations involving self-service technology, automation, and financial technology integration. The framework, methodologies, and findings presented in this research can guide them in

developing more advanced and efficient systems, particularly those that aim to improve public access and service automation in academic or institutional settings.

Scope and Limitation

This study focuses on the development and evaluation of a web-based system that automates the complete printing workflow within a library setting. The system is designed to accept document uploads only in PDF, DOCX, and standard image file formats, with a maximum file size of 25 MB per document. A fixed pricing scheme is embedded in the system: 2 pesos per page for Letter and A4 sized black and white prints, 5 pesos per page for Legal sized black and white prints, and a flat rate of 4 pesos per page for all colored prints regardless of paper size. Cashless payment processing is implemented exclusively through the Xendit API. The implementation and evaluation of the system are conducted at the SKSU Isulan Library, with the primary users being the library patrons. In line with the research design, respondents are selected using purposive sampling, targeting users who directly interact with the system. The system is evaluated based on functionality, reliability, and user convenience using User Acceptance Testing, system generated logs, and structured survey instruments.

To maintain feasibility and alignment with the study objectives, several limitations are imposed. The system is limited to a responsive web interface and does not include a dedicated mobile application. File support is restricted to the specified formats and size constraints, excluding larger or unsupported files. The pricing model is fixed and does not allow dynamic pricing or promotional

adjustments. The study utilizes existing or commercially available printing hardware and does not assess custom or specialized devices. Furthermore, the implementation is geographically confined to the SKSU Isulan Campus Library and is not extended to other SKSU campuses. Methodologically, the findings are limited to data collected from purposively selected users and may not be generalized beyond the study context.

Operational Definition of Terms

The following terms are operationally defined as used in this study:

- | | |
|-------------------------|---|
| Administrator | - Refers to the person responsible for managing, maintaining, and monitoring the kiosk system, including generating reports of printed pages and sales. |
| Cashless Payment | - Refers to the incorporation of electronic or digital payment systems into the kiosk, allowing users to complete transactions without using physical cash, specifically through GCash or similar mobile wallet services. |
| Database | - Refers to the structured data storage system that records transaction logs, payment histories, and usage reports for administrative monitoring and analysis. |

Functionality	- Refers to the ability of the system to perform its intended operations effectively, such as printing, payment processing, and data recording without errors or failures.
GCash	- Refers to a mobile wallet application in the Philippines that allows users to pay bills, send money, and make purchases online or in-store using their smartphones.
Xendit	- Refers to the payment gateway integrated into the kiosk system that facilitates secure and real-time transaction processing between GCash and the printing system.
Raspberry Pi 4	- Refers to the small, single-board computer that serves as the main processing unit of the self-service kiosk, managing both hardware and software operations.
Reliability	- Refers to the consistency and dependability of the system in performing its tasks, ensuring that printing and payment processes function accurately and continuously without frequent breakdowns or errors.
Self-Service	- Refers to a system or process that allows users to perform tasks or access services independently

without the need for assistance from staff or personnel.

Printing Kiosk

- Refers to an automated machine or station designed to provide on-demand printing services. It allows users to upload, preview, and print documents directly from the kiosk.

SKSU-Isulan Library

- Refers to the designated location within the Sultan Kudarat State University – Isulan Campus where the kiosk system is installed and utilized by students and staff.

System Interface

- Refers to the graphical layout and interactive components that enable users to communicate with the system, including display menus, prompts for uploading files and confirming payments

User Convenience

- Refers to the level of ease, accessibility, and satisfaction experienced by the users while interacting with the system, particularly in printing documents and making payments efficiently.

Data Analytics

- Refers to the process of collecting, analyzing, and interpreting data such as transaction logs, daily usage, and sales reports. It is used to monitor printing volume and generate summary reports to aid the administrator.

CHAPTER II

REVIEW OF RELATED LITERATURE

The review of related literature and studies will be covered in this chapter. By giving insight into the characteristics of people that are fundamentally necessary to the study's title, "Self Service Printing Kiosk in SKSU-Isulan Library with Cashless Integration via Gcash".

Raspberry Pi

The Raspberry Pi was selected as the main controller of the self-service printing kiosk due to its sufficient processing capability, Linux-based operating system support, and compatibility with peripheral devices and network-based applications. Its General Purpose Input/Output (GPIO) pins allow seamless integration with printers, touchscreens, and other kiosk hardware, making it suitable for automation and embedded system applications. Studies have shown that the Raspberry Pi is widely used in engineering and research environments requiring real-time data processing and system control. Thakur et al. (2023) highlighted its effective application in areas such as industrial automation and machine vision, demonstrating its capability to handle both data acquisition and on-board processing.

Furthermore, the Raspberry Pi's support for programming languages such as Python and system-level libraries enables efficient development of user

interfaces, printing services, and API-based payment verification. Rani and Singh (2022) emphasized that compared to traditional microcontrollers, the Raspberry Pi offers advantages in applications requiring networking, image processing, and cloud connectivity, as it operates as a full Linux-based computer while retaining hardware-level control. These characteristics make the Raspberry Pi an appropriate platform for implementing an automated printing kiosk that requires file processing, payment validation, and reliable system coordination.

Inkjet Printer

Inkjet printers are commonly used in academic and small-scale service environments due to their ability to produce high-quality text and image outputs at a relatively low cost. Their compatibility with computer-based printing systems and support for standard document formats make them suitable for automated printing applications. Studies indicate that inkjet printers provide sufficient resolution for academic documents and images while maintaining compact hardware requirements, which is ideal for kiosk-based installations (Hanna, 2022).

In the context of automated printing systems, inkjet printers support digital workflow integration by allowing direct control through operating systems and print management services such as CUPS. Their multifunction capability, including printing and scanning, further enhances operational efficiency by reducing the need for multiple devices. The reliability, affordability, and ease of maintenance of inkjet printers make them an appropriate choice for self-service printing kiosks.

intended for student use, where consistent print quality and system stability are essential.

Touch Screen Display

Touch screen displays have become a core component of modern digital systems due to their ability to support direct and intuitive user interaction. Sirois (2024) emphasized that touch-based interfaces enhance usability and efficiency by allowing users to interact directly with on-screen elements, eliminating the need for external input devices. This interaction model has been widely adopted in self-service kiosks and other automated systems because it reduces user effort and improves overall accessibility.

In self-service applications, touch screen displays contribute to improved user experience by enabling clear navigation, real-time feedback, and streamlined task execution. Their compatibility with contemporary operating systems allows seamless integration with graphical user interfaces used in automated kiosks. In the context of a self-service printing kiosk, the use of a touch screen display supports independent operation by guiding users through document selection, print preview, and payment confirmation, thereby minimizing reliance on staff assistance and enhancing system efficiency.

Quick Response (QR) Code

QR codes are widely used in digital systems to enable fast and contactless interaction between physical interfaces and online services. Hayes (December

2025) noted that QR-based technologies support efficient data access and transaction initiation by allowing users to scan codes using standard mobile devices, making them particularly suitable for payment verification and information retrieval applications. Their ability to encode various data types enables seamless integration with modern digital platforms, including cashless payment systems.

In automated service environments, QR codes enhance usability by simplifying user actions and reducing manual input requirements. Their adoption in digital payment and service kiosks demonstrates their effectiveness in improving transaction speed and user engagement. However, Hayes (December 2025) also emphasized that secure system design is necessary to mitigate risks associated with scanning unverified codes. In the proposed self-service printing kiosk, QR codes are utilized within a controlled system workflow to facilitate secure payment confirmation while maintaining user convenience and system reliability.

Common UNIX Printing System (CUPS)

The Common UNIX Printing System (CUPS) is widely implemented in Linux-based environments to manage print jobs, queues, and printer communication in automated systems. Its support for the Internet Printing Protocol (IPP) enables reliable handling of both local and networked printers, making it suitable for centralized print control applications (CUPS, 2023). Studies indicate that CUPS provides a stable and standardized framework for integrating printers with computing platforms that require continuous and unattended operation.

In automated service systems, CUPS facilitates efficient print workflow management by processing print requests, converting document formats, and transmitting data to printers without manual intervention. Its modular architecture allows seamless integration with application software and user interfaces, supporting real-time print job monitoring and control. In the proposed self-service printing kiosk, CUPS serves as the core printing service that ensures consistent print output, system reliability, and compatibility with the Linux-based controller used in the implementation (CUPS, 2023).

Xendit

Xendit is a digital payment gateway widely used in Southeast Asia to facilitate secure and efficient electronic transactions across multiple payment channels. Korostashevych (March 2025) emphasized that Xendit supports various payment methods, including e-wallets and bank transfers, through API-based integration that enables automated transaction processing and real-time payment verification. Its compliance with PCI DSS standards further ensures the security and integrity of financial data during digital transactions.

In automated service systems, payment gateways such as Xendit play a critical role in ensuring reliable transaction handling and system synchronization. The platform's ability to generate payment requests, confirm transaction status, and log payment records enables seamless integration with application-level services. In the proposed self-service printing kiosk, Xendit serves as the intermediary payment service that validates GCash transactions and ensures that

printing processes are executed only after successful payment confirmation, thereby improving transaction accuracy, security, and operational efficiency.

Gcash

GCash is a mobile-based digital wallet widely used in the Philippines to facilitate secure, cashless transactions. Afable (May 2024) notes that GCash supports payments, money transfers, bill settlements, and mobile commerce through a single integrated platform, enabling users to perform financial transactions efficiently and reliably. The application's regulatory compliance and security features, such as Customer Protect and QR code verification, ensure transaction integrity and user trust.

In automated payment systems, digital wallets like GCash serve as intermediaries that validate and authorize financial transactions in real time. Their integration via API allows seamless communication between the wallet and application-level services, ensuring that downstream processes are executed only after successful payment confirmation.

Self Service Technology

Self-service technologies (SSTs) are digital systems that allow users to independently perform transactions or access services without direct employee assistance. Malpass (2018) highlights that SSTs enhance operational efficiency by automating routine tasks while providing users with real-time access to services and information. Key features of effective SSTs include system integration,

multilingual support, and immediate data access, enabling seamless interaction between users and both digital and physical service environments.

In automated service systems, SSTs play a critical role in reducing manual intervention and ensuring reliable service delivery. For the proposed self-service printing kiosk, SSTs allow users to initiate printing tasks, select documents, and execute transactions without staff assistance. By integrating SSTs with digital payment services like GCash and Xendit, the kiosk ensures that printing operations proceed only after successful payment verification, improving user convenience, transaction accuracy, and overall operational efficiency.

Modern Academic Library Innovation

Academic literature positions innovation as a core function of libraries seeking to meet evolving user needs (Scott and McNamee, 2015; Walter and Lankes, 2015). This innovation often involves the introduction of new technologies, such as makerspaces and digital hubs, despite concerns related to cost and system complexity (Lehnen, 2019). Successful implementation is guided by frameworks such as design thinking, which emphasizes iterative and user centered development (Clarke, 2020), as well as strategies that promote user participation and external partnerships (Yeh and Walter, 2017). The proposed self-service printing station is situated within this context, representing a service innovation that renews a core library function by improving user convenience and operational efficiency through automation and cashless payment. This directly responds to the documented need for libraries to adapt and enhance their service models.

Cashless Payment

Cashless payments are transactions executed without physical currency, using electronic methods such as credit cards, debit cards, mobile wallets, bank transfers, and contactless technologies. Team (2025) emphasizes that these systems enhance transaction speed, convenience, and security through encryption, tokenization, and biometric authentication, ensuring reliable payment processing.

In automated service environments, cashless payment systems play a critical role in operational efficiency by streamlining transaction handling, reducing processing delays, and maintaining accurate payment records. They also contribute to financial inclusion by providing access to digital financial services for individuals without traditional banking accounts. In the proposed self-service printing kiosk, cashless payments serve as the primary method for executing transactions. Integration with platforms such as GCash and Xendit ensures that print services are activated only after successful payment confirmation, improving reliability, security, and user convenience.

Self-Service Technology (SST) in Educational Institutions

According to Bell (2022), in *Moving to Mobile: Space as a Service in the Academic Library*, self-service technologies (SSTs) have increasingly been adopted by educational institutions to improve service efficiency, reduce queues, and provide students with greater autonomy. One notable example discussed is the installation of laptop and battery vending kiosks at Charles Library at Temple

University. These kiosks allow students to borrow computing devices in a self-service manner, enhancing access particularly for students who do not regularly bring laptops or whose devices may be temporarily unavailable. Another study by Wendy S. Wilmoth (2020), in *Circulating Laptops in a Two-Year Academic Library: A Formative Assessment*, examined a laptop loan program at Georgia Piedmont Technical College. The study evaluated student satisfaction, identified logistical challenges, and analyzed how circulation policies influenced laptop usage, providing valuable insights for improving library technology services.

In the Philippines, the study of Carretero, & Apolinario (2022), SSTs are also gaining ground in academic libraries. For instance, the STARBOOKS (Science and Technology Academic and Research-Based Openly-Operated Kiosks) program was evaluated based on user perception” (UP Diliman) that examined how users perceive and interact with these kiosks to access digital resources. Also, Bukidnon State University (BukSU), (2023) recently introduced a Self-Check Kiosk for its library, aiming to ease borrowing and returning books by allowing users to perform these tasks without assistance, which was reported to improve convenience for students and staff.

Cashless Payment Adaptation and User Acceptance in the Philippines

The adoption of cashless payment systems in the Philippines has accelerated rapidly, particularly following the COVID-19 pandemic. Recent studies indicate a significant shift in consumer behavior, with Visa's Consumer Payment Attitudes Study reporting that 84% of Filipino consumers attempted cashless

transactions in 2021, with mobile wallets emerging as the most preferred channel at 64% (Luz, 2022). This trend continued into 2022, with 82% of Filipinos attempting cashless payments and 50% carrying less physical cash as digital options became more accessible (Navarro, 2023). The mobile wallet GCash has been central to this transformation, with studies demonstrating high user satisfaction due to its convenience, security features, and customer service (Montenegro, Afana & Alico, 2022). Research applying the Technology Acceptance Model (TAM) further supports this trend, identifying perceived ease of use, perceived usefulness, and trust as key factors driving e-wallet adoption among younger demographics in the Philippines (Belmonte et al., 2024). This widespread acceptance and positive user perception provides a strong foundation for integrating GCash into self-service systems like printing kiosks in academic environments.

Operational and Institutional Benefits of Cashless Systems

The implementation of cashless systems offers significant operational advantages across various sectors. Studies demonstrate that cashless payment adoption reduces costs associated with handling physical cash while increasing transaction speed and accuracy (Rahman et al., 2022). These benefits extend to institutional settings, where digital payments streamline processes and reduce administrative workload. In the Philippines, this trend is evident in public service integrations, such as GCash partnerships with local government units for tax payments and social welfare distribution, which have improved efficiency and

reduced physical queues (Gabay, 2021; Susilo & Dizon, 2023). Within educational contexts, similar benefits can be realized through self-service kiosks, where cashless transactions minimize manual payment handling, accelerate service delivery, and provide automated transaction records for monitoring and reporting. These operational efficiencies directly align with the goals of implementing a cashless printing kiosk in a university library setting.

Kiosk-Based Automation and System Architecture

Self-service kiosk architecture that integrates backend monitoring, notifications, and user interaction features has been explored in academic projects to improve usability and efficiency. For example, “Optimizing User Interaction and Notification Efficiency in Smart Coin-Operated Printing Kiosk with Real-Time SMS Notifications and Interactive Features” by Geneveve et al., (2024), describes a prototype coin-operated printing kiosk with a system architecture that includes error detection (via GSM/SMS), interactive features, and a monitoring system to track user interactions. The study evaluates accuracy, effectiveness, and learnability, showing the system is usable and efficient in a campus setting. However, “Design and Development of an Information Kiosk with Log Monitoring for Leonor M. Bautista National High School”, it presents an information kiosk built with log-monitoring to record usage (who accessed, when, what info), and an architecture that supports dissemination of information while tracking visitor interactions. This helps the institution to monitor both the performance and usage patterns of the kiosk (Bulaclac et al., 2023).

AI and IoT-Enhanced Self-Service Kiosks

Based on Peña et al. (2023), integrates IoT, AI, and remote monitoring/services in kiosk systems to enable automation and real-time control. “An AI-integrated IoT-based Self-Service Laundry Kiosk with Mobile Application” describes “KILAO”, a self-service laundry kiosk. The system architecture combines hardware (laundry machines), IoT sensors, a mobile app, queue management via AI, and supports various payment options (cash, cashless, card). It demonstrates how combining backend cloud connectivity, AI prediction of queue/wait times, and user interface through a mobile app enhances both operational efficiency and user convenience. This one focus more on UX vs architectural/performance expectations: it compares what customers expect of a kiosk system (in terms of responsiveness, reliability, interface delays) and what they actually experience. Though less heavy on hardware architecture, it highlights system performance metrics such as processing speed, error rate, interface design, and how they are integral to a good kiosk architecture (Zhari et al., 2023).

Cybersecurity in Cashless Systems

As cashless systems proliferate, cybersecurity threats become more prominent, affecting user trust and system integrity. One study, “A Study on Cyber Attacks and Vulnerabilities in Mobile Payment Applications” Rivers et al., (2020), analyzes common mobile payment platforms like Apple Pay, Google Pay, Samsung Pay; it identifies points of vulnerability in authentication, authorization,

verification chains, and transaction processing, which surveys 50 fintech / bank mobile payment apps in both high and low technological-advancement countries. It evaluates vulnerabilities per the OWASP / CWE standards, including issues related to cryptography, certificate handling, and insecure communications (Archibong et al., 2024).

Data Privacy and Protection in FinTech and E-Wallet Services

According to Npbetia (2021), data privacy is also a key concern, especially with regard to how user data is collected, stored, and processed, which can deter adoption of cashless services. For example, “Data Privacy Issue of Mobile Banking Users and Other E-Wallet Servicing Applications in the City of Bacolod” examine the concerns users have about personal information safety, usefulness of apps, and why some remain hesitant to fully use mobile-financial services due to perceived privacy risks. However, “Big Data Privacy in Emerging Market Fintech and Financial Services: *A Research Agenda*” which does not limit itself to one country but outlines priority areas: how fintech services in emerging markets gather, protect, and use large datasets, and discusses technical & policy-solutions such as differential privacy, encryption methods, and legal/regulatory oversight (Blumenstock & Kohli, 2023).

User Experience (UX), Convenience, and Accessibility in Digital Transactions

Digital transactions user experience is heavily influenced by how easy and useful people perceive e-wallets to be. One study titled “Factors Influencing the Intention to Use E-wallet among Generation Z and Millennials in the Philippines: An Extended Technology Acceptance Model (TAM) Approach” found that perceived ease of use, perceived usefulness, and perceived trust are among the strongest predictors of intention to use e-wallets among younger generations (Belmonte et al., 2024). In addition, “Effectiveness of E-wallet Transactions in a Small Business Enterprise in Batasan Hills, Quezon City: Developing a Guidelines” examined small business settings and found that convenience, payment accessibility, and portability significantly enhance users’ satisfaction with e-wallet transactions; issues like unreliable internet and app functionality negatively affect user experience (Tecson et al., 2023).

Demographic Influences on E-Wallet Adoption and User Experience

Accessibility and the broader impact of digital wallets on different demographic groups also play a big role. In “Digital wallets usage in the Philippines: determining the impact of demographics on complementarity versus substitutability, Reyes and Tenedero, (2021), investigated whether people view e-wallets as complementary to bank accounts or as substitutes, and how age, gender, employment affect that. They found that demographics significantly influence how users adopt and integrate digital wallets into their lives, which has implications for design (making interfaces and functions accessible for different user groups). Moreover “Examining consumers attitude towards E-wallet

utilization as a payment method using technology acceptance model: A perspective from young generations in Nueva Ecija, Philippines” looks at UX and perceived risks (like privacy, security), showing that ensuring clear feedback, trust signals, and reliable performance are important for user convenience and continued usage (Leyesa et al., 2024).

Self-Service and Automation Initiatives in Mindanao

In Mindanao, several academic institutions have started adopting self-service and automated systems to improve operational efficiency and user accessibility. For instance, Bukidnon State University (BukSU), (2023), introduced a self-check kiosk for library transactions, enabling students to borrow and return books independently without librarian assistance. This innovation was designed to reduce waiting time and promote digital literacy among library users. Similarly, a study conducted at the University of Mindanao in Davao City explored the use of self-service kiosks in fast-food restaurants, revealing that customers highly value systems that offer convenience, speed, and reliability in transactions. The study concluded that user satisfaction is strongly influenced by the simplicity of navigation and the integration of cashless payment features (Del, Rojo & Marie, 2024).

Acopiado et al. (2022) from UP Mindanao conducted a study on digital payment adoption during the COVID-19 pandemic. They found that adoption in Mindanao businesses was influenced by infrastructure, firm size, and digital literacy. Their study suggests that regions outside Metro Manila face unique

barriers such as weaker internet infrastructure, which must be considered when implementing kiosks in libraries.

Conceptual Framework

This study was anchored on the Input–Process–Output (IPO) model, which served as the conceptual guide in developing the Self-Service Printing Kiosk in SKSU–Isulan Library with Cashless Payment Integration via GCash. The framework illustrates how the different components of the system interact—from the required inputs, to the processes of design and development, and finally to the expected outputs and feedback.

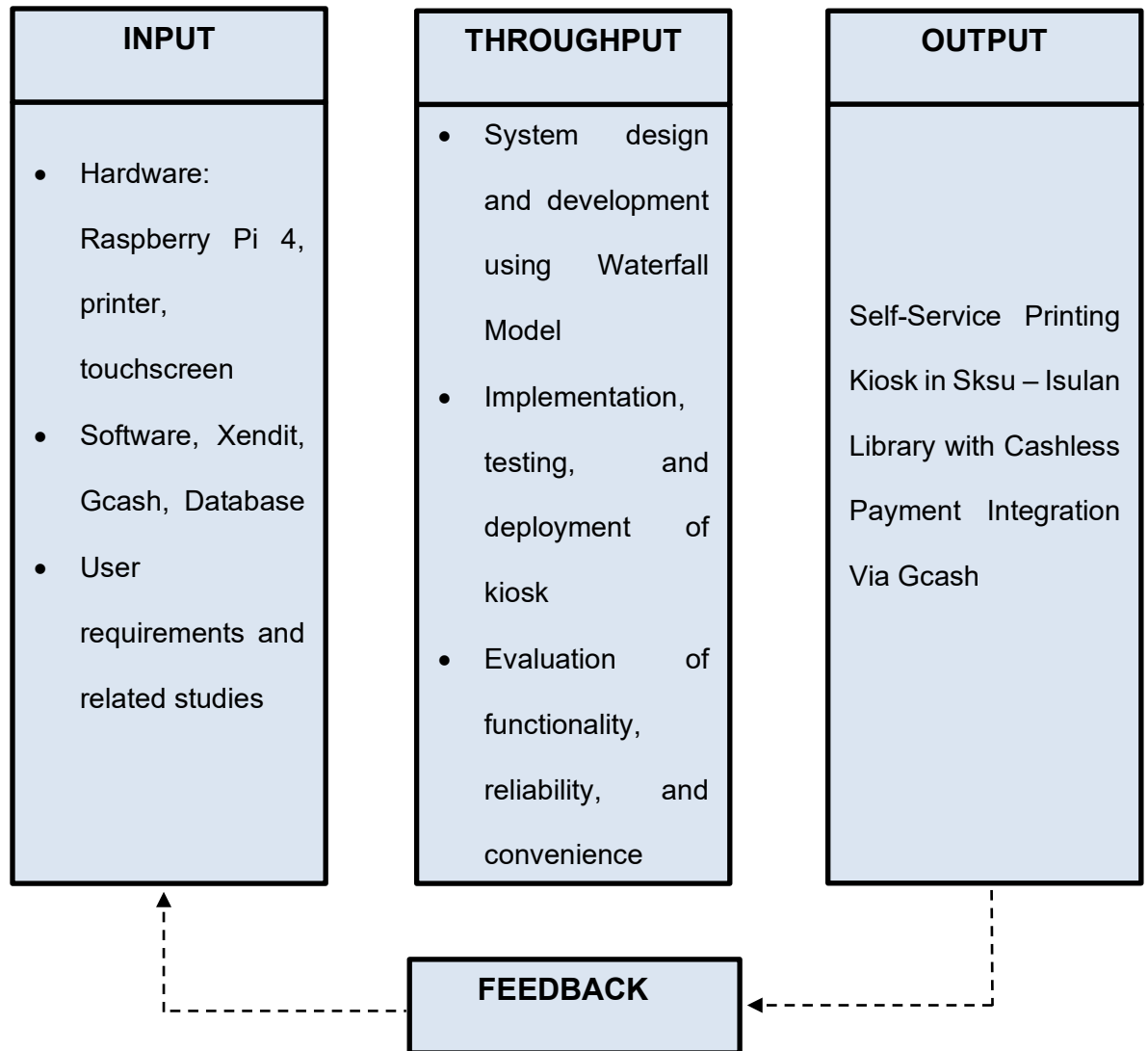


Figure 1. Conceptual Framework of the Study.

Input

The input phase includes all the resources, information, and requirements necessary for the system's development. These consist of the hardware and software components such as the Raspberry Pi 4, touchscreen monitor, inkjet printer, Xendit API integration and Gcash. The gathered data on system requirements, user needs, and existing printing practices within the SKSU–Isulan Library are also part of this stage. In addition, the conceptual and technical

references reviewed in the related literature provide theoretical and empirical support for the system's design.

Process

The process phase represents the actual development, implementation, and testing of the system using the Waterfall Model of the System Development Life Cycle (SDLC). This stage involves:

1. **Requirements Analysis** – identifying user needs and system functions such as file uploading, page counting, and payment verification;
2. **System Design** – defining the hardware and software architecture of the kiosk;
3. **Implementation** – integrating the GCash payment gateway and user interface through programming and hardware configuration;
4. **Testing** – assessing the system's performance in terms of functionality, reliability, and user convenience; and
5. **Deployment and Maintenance** – installing the kiosk in the SKSU–Isulan Library for real-world use and preparing it for future upgrades.

Output

The output phase produces the fully operational Self-Service Printing Kiosk, capable of providing an automated, cashless, and user-friendly printing service. The system allows students and library users to upload files, pay through GCash,

and automatically receive printed documents without staff assistance. It also generates weekly and monthly reports for the administrator to monitor sales and usage statistics.

Feedback

The feedback phase involves collecting and analyzing user evaluations to determine the system's effectiveness and reliability. Feedback from students, library staff, and administrators provides insights for improvement and serves as a basis for future enhancements of the kiosk, including interface design, system speed, and additional payment options.

CHAPTER III

METHODOLOGY

This chapter presents the Hardware Requirement used in the study, Bill of Supply and Materials, Project Duration, System Development Life Cycle, Context Diagram, Data Flow Diagram, Hardware Development and Methodology, and Perspective Plan.

Project Development Description

Tools and Equipment

These tools and equipment presented are the equipment used and needed for the completion of the project.

Table 1. Tools and Equipment

Quantity	Unit	Tool/Equipment
1	Pc.	Saw
1	Pc.	Measuring tools
1	Pc.	Computer
1	Set	Pliers/Cutters
1	Pc.	Soldering Iron with solder lead
1	Pc.	Multimeter
1	Set	Screw drivers
1	Pc.	Electric Drill
1	Pc.	Heat Gun

Table 1 showed the device's tools and equipment used in constructing the Self-Service Printing Kiosk in SKSU – Isulan Library with Cashless Payment Integration via GCash. The saw was used for cutting acrylic, or metal sheets into the required dimensions for the kiosk enclosure. Measuring tools such as steel tape, rulers, and calipers were purposely used for accurate measurement and alignment during the construction of the kiosk frame and component placement. Wires were essential in establishing electrical connections between the Raspberry Pi, inkjet printer, power supply, and other electronic modules. Pliers and cutters were utilized for bending, gripping, and cutting wires during assembly. The soldering iron with solder lead was employed to solder and permanently connect electronic components and wires. A multimeter was used to test electrical connections, measure voltage, and verify continuity to ensure proper wiring and safety of the system. Screwdrivers, both flat and Philips types, were used for fastening and securing components such as the Raspberry Pi, power modules, and touchscreen monitor inside the kiosk. The electric drill was utilized to drill holes for mounting screws, switches, and ventilation slots in the kiosk frame. Lastly, the heat gun was used for applying heat on shrink tubes to insulate exposed wires and for proper cable management.

Table 2. Bill of Supplies and Materials

Quantity	Unit	Item description	Unit Price (PHP)	Total (Php)
1	pc	Raspberry Pi 4 (4GB) with 32GB microSD card	4,000.00	4,000.00
1	pc	Official Raspberry Pi Power Supply (5V 3A)	700.00	700.00
1	pc	Touchscreen Monitor (10–15 inch)	6,000.00	6,000.00
1	pc	Multi-function Printer (Inkjet/Laser)	8,000.00	8,000.00
5	pcs	Shrinkable Tube (for insulation)	30.00	150.00
5	m	Cable Duct / Cable Tie Set	25.00	125.00
1	pc	Waterproof Push Button Switch	120.00	120.00
2	m	Flat Cord #16	40.00	80.00
1	pc	Ethernet Cable / WiFi Module	500.00	500.00
1	set	Kiosk Enclosure (metal/plywood with acrylic)	2,000	2,000
1	set	Hinges, Locks, Bolts, Nuts & Screws	500.00	500.00
1	set	Ventilation Grills + Cooling Fan	600.00	600.00
2	pcs	Paint (for finishing)	250.00	500.00
SUB TOTAL				Php 23,275.00
Contingency (10%)				Php 2,327.50
Grand Total				Php 25,602.50

Shown in Table 2 are the bills of supply and material used in creating the Self-Service Printing Kiosk in Sksu – Isulan Library with Cashless Payment Integration Via Gcash including quantity, unit price, item description, and unit price, as well as the overall cost of the project design and the materials used to create the evaluation sheets, were displayed.

Project duration

Figure 2. Project Duration

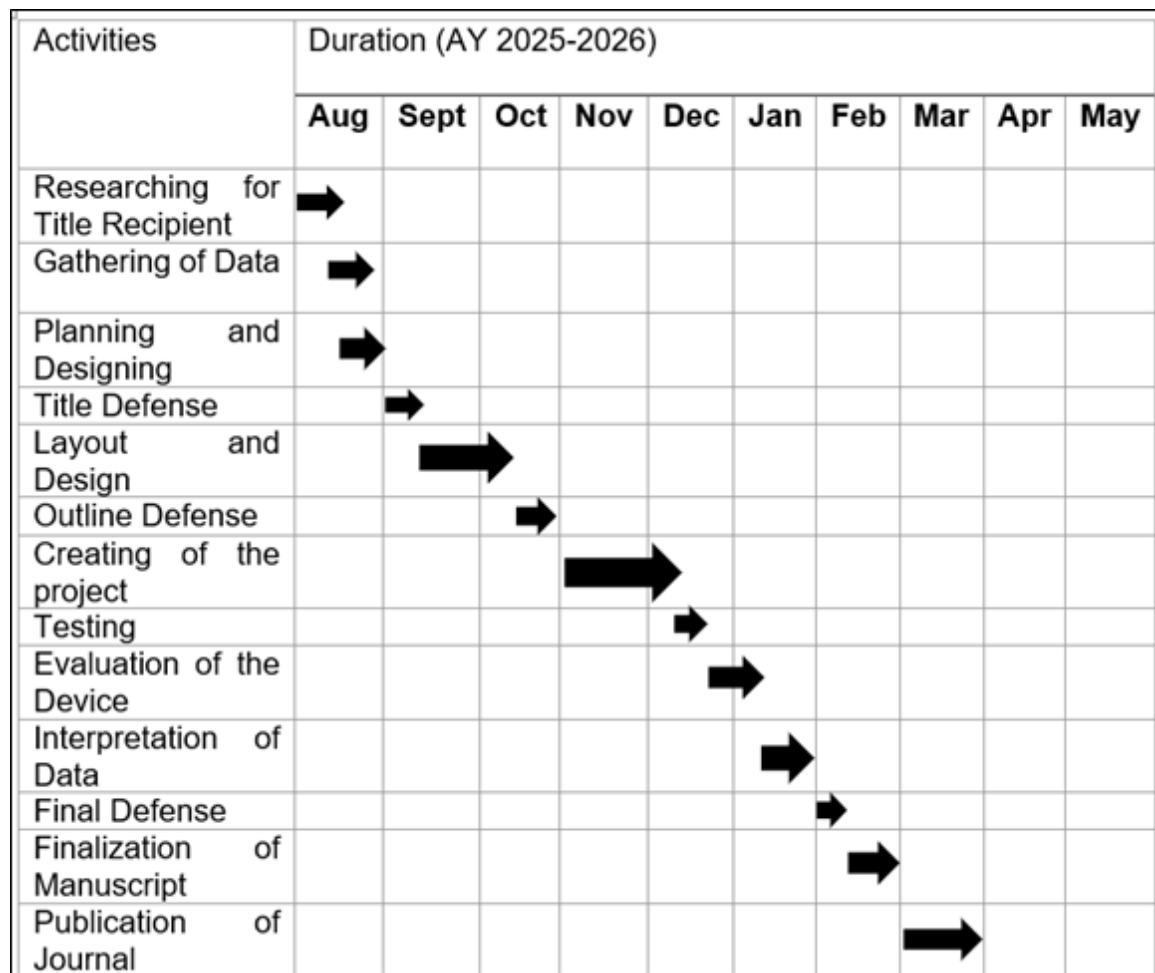


Figure 2 shows the project duration of the Self-Service Printing Kiosk in SKSU – Isulan Library with Cashless Payment Integration via GCash. As indicated, the conceptualization for the title proposal started in the first week of August 2025 until the end of the month. The title defense was conducted in the first week of September 2025. The gathering of tools and equipment took place in September 2025 until first week and second week of October, followed by the outline defense in October 2025.

Software Development Methodology

The software development methodology will adhere to the waterfall methodology model depicted below:

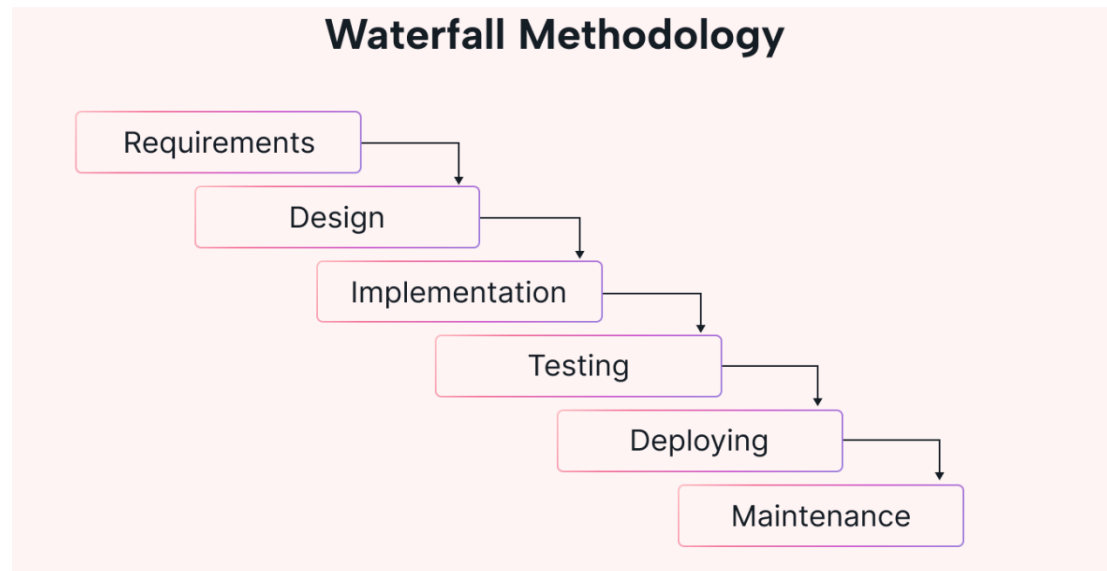


Figure 3. System Development Life Cycle

The researchers began by identifying the problems and challenges encountered by students in accessing affordable, efficient, and self-service printing facilities within the library. A project title was then formulated, together with the objectives, scope, and limitations of the study. Issues that arose during the design and development of the system were also recognized and addressed throughout the process of the study entitled Self Service Printing Kiosk in SKSU – Isulan Library with Cashless Payment Integration via GCash.

The **Waterfall Model** was selected as the development approach since it follows a structured and linear framework with clearly defined phases. This model ensures that each stage of the development is completed before proceeding to the next, making it suitable for a system that requires well-documented processes and

minimal iterations. The phases of the Waterfall Model applied in the study are presented as follows:

Requirements Analysis

The researchers gathered and analyzed the functional requirements of the kiosk. This included file uploading, page counting, Xendit API, payment verification, and printing. Non-functional requirements, such as user-friendliness, reliability, and system security, were also identified and documented.

System Design

In the design phase, the hardware and software components were mapped and specified. The Raspberry Pi was chosen as the main controller, integrated with the printer, and touchscreen monitor. On the software side, CUPS was configured for printer management, Xendit API was designed for cashless transactions, and a database structure was outlined for transaction logging and reporting.

Implementation

The development phase involved programming the kiosk interface using Python and web technologies, integrating the printer drivers, and linking the Xendit API. The Raspberry Pi was set up with the required operating system and software packages. Hardware components were also interfaced with the software to enable communication and automation.

Testing

Testing was conducted to evaluate the reliability and performance of the kiosk. Each module was tested individually, such as file uploading, payment verification, and printing, followed by system-level testing. Trial-and-error

procedures were carried out to identify bugs and errors, which were resolved before moving to the next stage.

Deployment and Maintenance

The finalized kiosk was deployed in the SKSU – Isulan Library for actual use. During this stage, final adjustments were made to improve user interaction, transaction speed, and system security. Maintenance provisions were included to allow future updates, bug fixes, and integration of additional features if needed.

Context Diagram

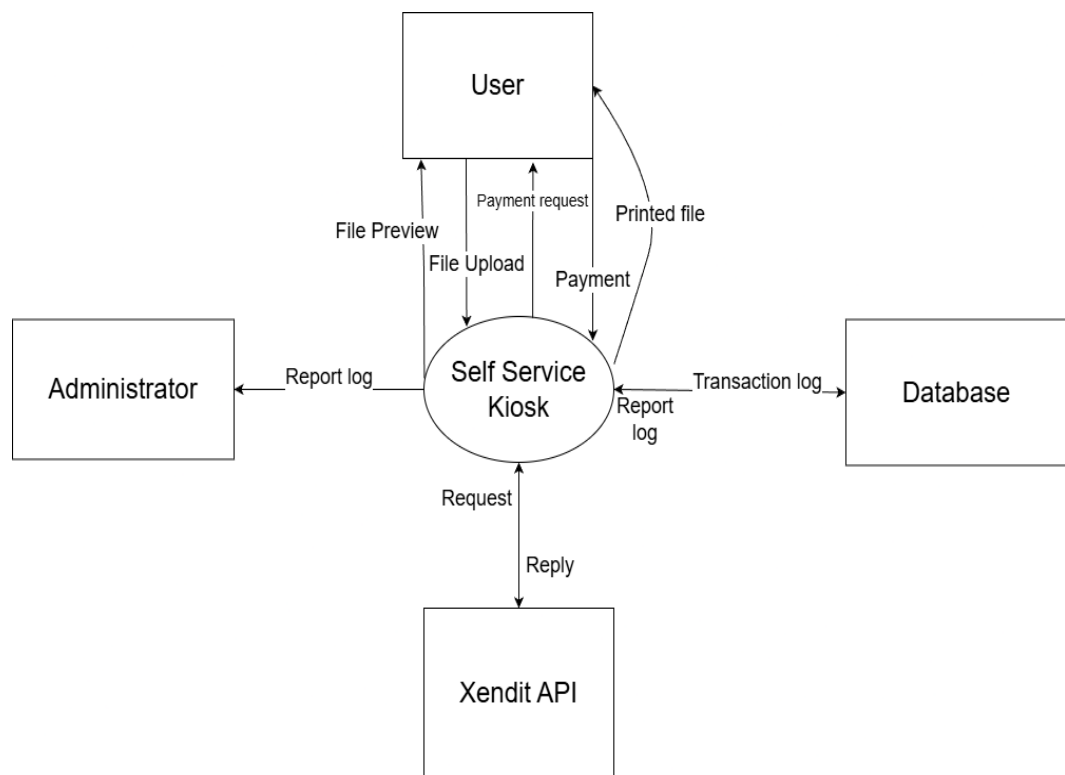


Figure 4. Context Diagram of the System

Figure 4 shows the context diagram of the Self-Service Printing Kiosk in SKSU – Isulan Library with Cashless Payment Integration via GCash. It illustrates

the system as a single process that interacts with external entities such as the User, Administrator, Xendit API, and the Database.

The diagram demonstrates the flow of inputs and outputs. The User uploads a file via qr code, previews the file, Xendit API for payment, generate qr code for payment, and receives the printed document as output. The Xendit API processes the payment request sent by the kiosk and provides confirmation of successful transactions. The Administrator receives weekly and monthly log reports of sales and printed pages for monitoring purposes. The Database stores transaction records, log files, and payment information, while also sending feedback on completed printing jobs.

The purpose of the diagram is to emphasize the boundaries of the kiosk system and how it exchanges inputs and outputs with its external actors without detailing the internal processes.

Data Flow Diagram

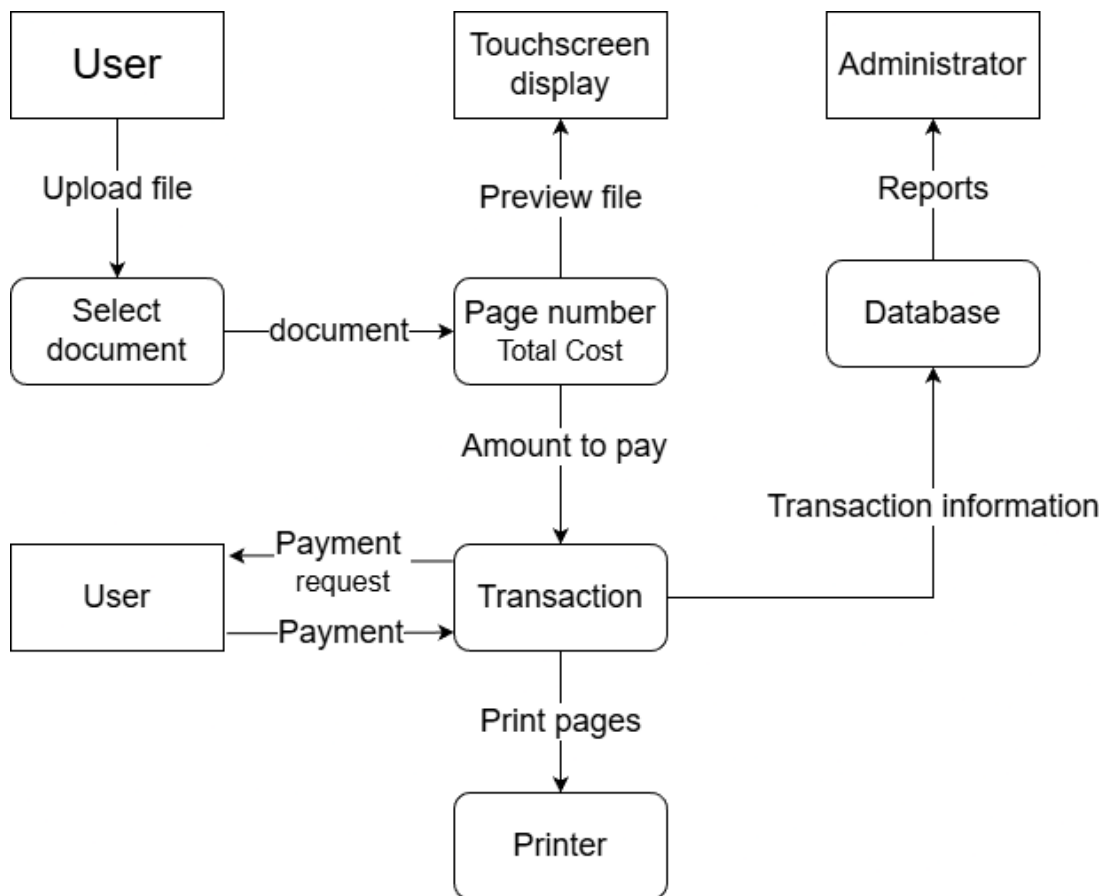


Figure 5. Data Flow Diagram

The figure 5 shows the Data Flow Diagram of the Self-Service Printing Kiosk in SKSU – Isulan Library with Cashless Payment Integration via GCash. It illustrates the breakdown of the system into processes and demonstrates how data moves between the kiosk and its external entities.

The User uploads a file to the kiosk system, which is automatically undergoes page counting to determine the cost of printing. The system then sends a payment request to the Xendit API and then link Gcash for payment. Once

payment is confirmed, the kiosk forwards the file to the Printer, which produces the printed document that is received by the user.

Meanwhile, the system records all transaction details and logs into the Database, which stores information on pages printed, payments received. The Administrator receives weekly and monthly reports generated from these logs to monitor the kiosk's performance and sales.

The purpose of the Data Flow Diagram is to show the interaction of the kiosk's internal processes with its external entities, highlighting how data inputs (files, payments) are transformed into outputs (printed documents, reports, confirmations).

Entity Relationship Diagram

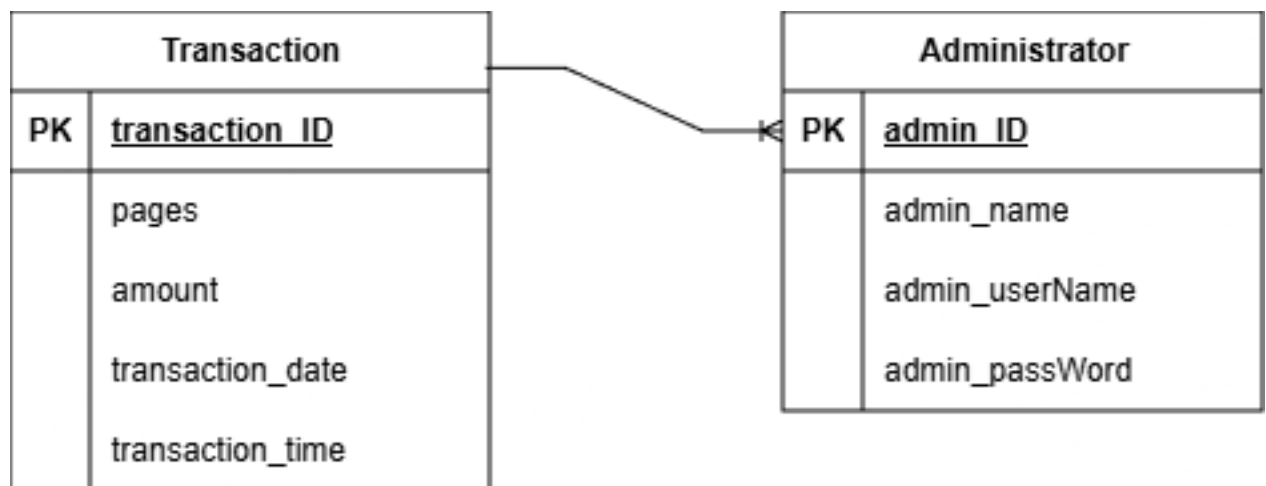


Figure 6. Entity Relationship Diagram

Figure 6 shows the Entity-Relationship Diagram (ERD) of the Self-Service Printing Kiosk System. The diagram illustrates the two primary entities Administrator and Transaction and the relationship between them. The

Administrator entity stores information about the authorized personnel responsible for monitoring and managing the kiosk operations.

The Transaction entity records all printing activities performed on the kiosk, including details such as the file page count, amount, transaction date, and transaction time. Each transaction is linked to a specific administrator, forming a one-to-many relationship, where one administrator can oversee multiple transactions.

This structure ensures organized data management within the system by maintaining clear accountability over printing activities and enabling efficient monitoring, reporting, and auditing of kiosk usage.

Database Structure

Table 3. Administrator Table

Field Name	Data type	Size
admin_name	varchar	50
admin_id	Int	11
admin_userName	varchar	50
admin_passWord	varchar	50

Table 3 shows the admin account data structure of the system, including the admin's information, such as name username, and password. The administrator account is stored in this table for storage purposes.

Table 4. Transaction Table

Field Name	Data Type	Size
transaction_id	INT	11
total_pages	INT	11
amount	INT	5
transaction_date	DATE	
transaction_time	TIME	

Table 4 shows the transaction history table, which stores records of all printing transactions made through the self-service printing kiosk. Each transaction has a unique ID that helps identify and track specific printing activities. The table includes information such as the user's ID, total number of pages printed, total amount paid, and the corresponding date and time of the transaction. This table is essential for maintaining accurate records of printing usage, enabling efficient monitoring, management, and financial tracking of the kiosk system.

Hardware Development Methodology

The hardware development of the Self-Service Printing Kiosk in SKSU – Isulan Library with Cashless Payment Integration via GCash will follow a systematic approach to ensure that all the physical components are properly designed, constructed, and integrated with the software system. This methodology consists of three main parts: Perspective Plan, Construction Procedures, and Block Diagram. These components ensure that the design process is organized,

functional, and aligned with the system requirements. Each stage focuses on achieving accuracy in hardware configuration and establishing a strong foundation for the successful operation of the system.

Perspective Plan

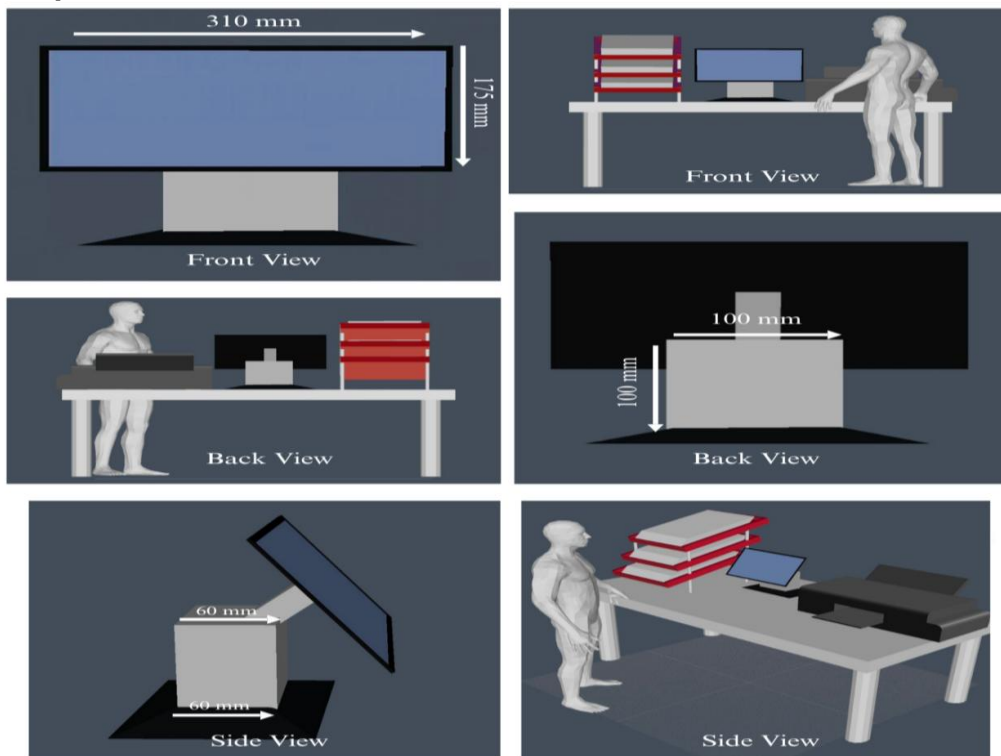


Figure 6. Perspective Plan of the Device

Figure 6 shows the Perspective Plan of the Self-Service Printing Kiosk in SKSU – Isulan Library with Cashless Payment Integration via GCash, illustrating the physical layout and design of the system. The display used in the setup measures 14 inches with a 16:9 aspect ratio, equivalent to 310 mm in width and 175 mm in height. The Raspberry Pi unit is enclosed in a protective case measuring 100 mm in height, 100 mm in length, and 60 mm in width. This configuration provides a compact and organized arrangement, ensuring efficient

interaction between the user interface, payment module, and printing components within the kiosk system.

Construction Procedures

The construction processes for developing the project were as follows:

1. Gather all materials, supplies, and equipment required for constructing the kiosk, including the Raspberry Pi, touchscreen monitor, inkjet printer, power supply, wiring, cooling fan, enclosure materials, hinges, and locks.
2. Fabricate the kiosk enclosure, which serves as the housing for all internal components such as the Raspberry Pi, power supply, and cable management system. Ventilation grills and cooling fans were also considered in the design to prevent overheating.
3. Install the main components, including the inkjet printer in the side section of the kiosk for document output, the touchscreen monitors at eye level for the user interface, the Raspberry Pi on its mounting bracket.
4. Program the Raspberry Pi with the appropriate code to manage the file upload, page counting, Xendit API, payment request handling, printer control, and database logging.
5. Connect the electrical and electronic components, including the Raspberry Pi GPIO pins, printer interface, and cooling system, ensuring proper circuit assembly and insulation with shrink tubes and cable ducts.

6. Conduct initial testing of all components to verify if the Raspberry Pi boots properly, the printer responds to print requests, the touchscreen displays the interface, the Xendit API detects payments, and the payment confirmation from GCash allows the print job to proceed.
7. Perform troubleshooting and final adjustments based on the test results. Retest the system to ensure complete functionality, safety, and reliability of the kiosk before deployment in the SKSU – Isulan Library.

Flowchart

Figure 7 illustrates the flowchart of the functionality of the Self-Service Printing Kiosk System. The process begins when the power system is activated, supplying energy to the Raspberry Pi, touchscreen monitor, and printer. Users upload their files through the interface, which are then received and processed by the Raspberry Pi. The user selects a file, and automatically counting the number of pages. The file is then displayed on the monitor for user verification.

If the user decides to proceed with printing, the system redirects to Gcash for payment. Once payment is confirmed, the printing process begins. However, if the user cancels the printing request or if the payment is not verified, the system terminates the process and returns to standby mode. The operation concludes when the printing is completed or the transaction is canceled, ready for the next user

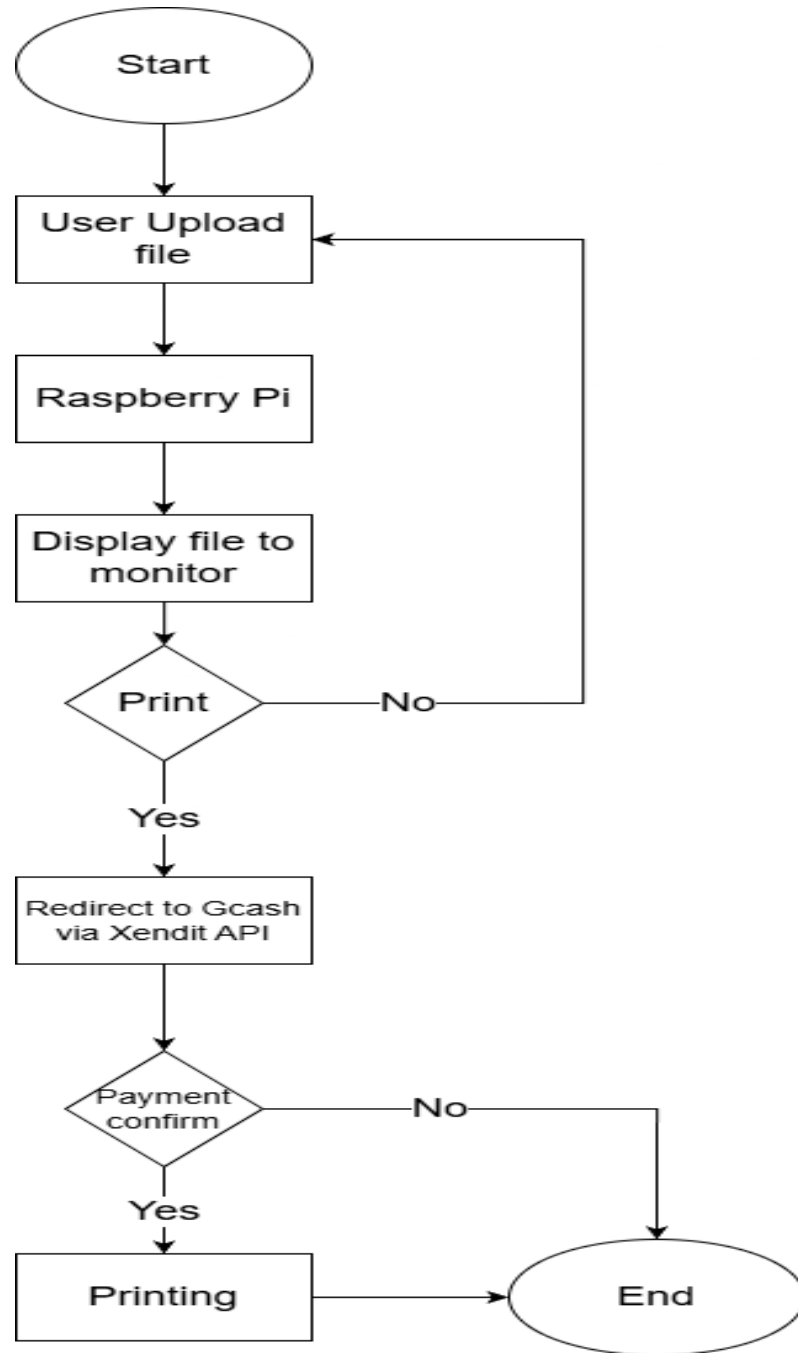


Figure 7. Flowchart

Evaluation Methodology

Research Design

The type of method that was used for this study was descriptive research. The researchers are required to gather information from the respondents by providing a questionnaire for evaluation of the study. The questions from the questionnaire evaluate the functionality, reliability and user convenience of the self-service printing kiosk.

Locale

Figure 8 shows the locale of the study, which will be conducted at Sultan Kudarat State University, Isulan Campus, specifically in the Isulan Library. The location was chosen because it is frequently visited by students who need to print their academic requirements, such as research papers, assignments, and reports. The library provides an ideal environment for implementing and testing the Self-Service Printing Kiosk with Cashless Payment Integration via GCash, as it directly addresses the common issues of long queues, manual payment processes, and limited printing accessibility. Conducting the study in this setting allows researchers to evaluate the kiosk's effectiveness in improving printing efficiency, promoting cashless transactions, and enhancing the overall convenience and productivity of students and library staff.

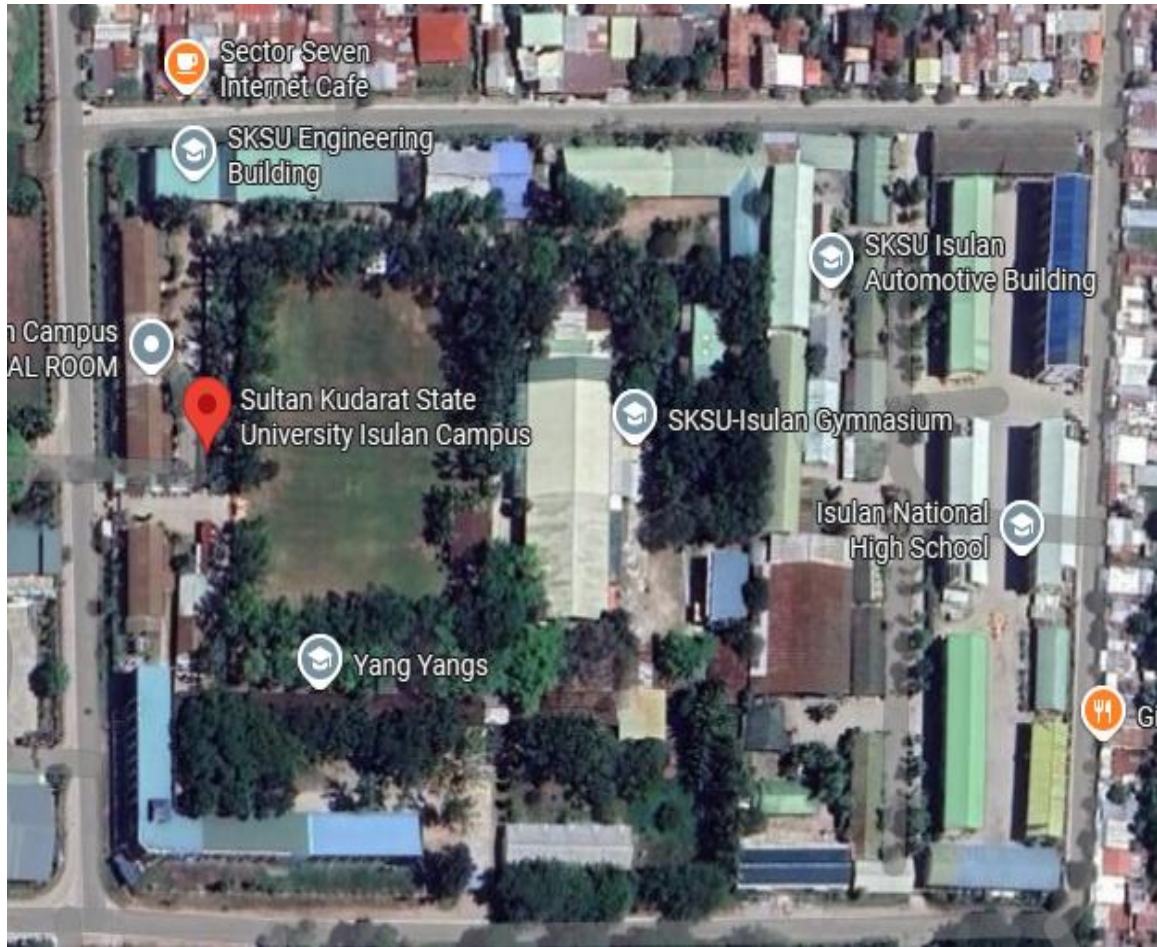


Figure 8. Satellite map of Sultan Kudarat State University Isulan Campus

Methods of Research

To fulfill the objectives of the study, a descriptive survey will be used. The creation of a study questionnaire is the first action taken by researchers. After the questionnaire has been created and distributed to respondents, the self-service printing kiosk are evaluated. In the descriptive survey that was conducted, which will serve as the foundation for determining how reliable, effective and efficient the said system is, the respondents' perspectives, experiences, and views regarding the mobile application built are addressed.

Respondents of the Study

The respondents of this study will be composed of the first 200 users which would either be students from the College of Engineering, College of Computer Studies and College of Industrial Technology, faculty members, and staff of Sultan Kudarat State University – Isulan Campus, who are regular users of the university library and its facilities. A total of two hundred (200) respondents will be randomly selected to participate in the evaluation of the developed self-service printing kiosk. The inclusion of different groups of respondents ensured that the system was assessed from multiple perspectives — as end users, educators, and administrative personnel.

The respondents were chosen through random sampling to ensure fairness and representativeness of opinions from various departments within the university. The diversity of the respondents allowed for a comprehensive and balanced evaluation of the self-service printing kiosk, ensuring that the results of the study accurately reflect the system's overall performance and user satisfaction across different user groups.

Data Gathering Procedures

The researchers sought permission and authorization from the University faculty and staff. The researchers surveyed to gather the information that would be deemed helpful for the study. Throughout the survey, the researchers guaranteed that the respondents' identities were treated with the utmost confidentiality.

The following are the procedures on how the research gathers all the data and information regarding the project.

1. Prepare the web application and administrator website for evaluation.
2. Making research questionnaire.
3. Finalization of the questionnaire.
4. Distribution of questionnaire.
5. Testing and actual evaluation of the application.

Data Gathering Instruments

This study used a survey questionnaire as the data-gathering instrument. The instrument was chosen since it is cost-effective and easy to manage. The questionnaire comprises related questions wherein the respondents will rate the self-service printing kiosk based on their experience. The researcher used a hard copy of the evaluation form to gather the data.

Statistical Tools and Treatment of Data

After evaluating the Self-Service Printing Kiosk in Sksu – Isulan Library with Cashless Payment Integration Via Gcash, the researchers tallied the evaluation results and collected the respondents' evaluation forms. The respondents' responses served as the basis for data tabulation.

Data analysis and interpretation were done using descriptive statistics such as the 5-point Likert Rating Scale. The 5-point Likert Rating Scale was statistically

treated to determine the application's functionality, reliability, and user convenience. The 5-point Likert Scale will be computed using the formula below.

$$WM = \frac{1(SD) + 2(d) + 3(MA) + 4(A) + 5(SA)}{n}$$

Where: **WM** = Weighted Mean

n = number of respondents

SD = Strongly Disagree

D = Disagree

MA = Moderately Agree

A = Agree

SA = Strongly Agree

Table 5. Interpretative Scale used to Interpret the Mean

Numerical Scale	Interpretation
4.20 - 5.0	Strongly Agree
3.40 - 4.19	Agree
2.60 - 3.39	Moderately Agree
1.80 - 2.59	Disagree
1.00 - 1.79	Strongly Disagree

Table 5 displays the application's numerical ratings based on its overall mean. It demonstrates that the device Strongly Agree when its overall mean falls between 4.20 and 5.00. After that, the gadget receives descriptive ratings of Agree when its overall mean ranges from 3.40 to 4.19 and Moderately Agree when it ranges from 2.60 to 3.39. Additionally, the device

will receive a descriptive rating of Disagree when its overall mean is between 1.80 and 2.59 and Strongly Disagree when its overall mean is between 1.00 and 1.79.

Table 6. Rating Scale used in Evaluating the Functionality of the System.

Rating	Interpretation
5	Highly functional
4	Functional
3	Moderately functional
2	Less functional
1	Not functional

Table 6 shows the rating scale and interpretation as a reference to evaluate the system functionality. A rating of 5 is considered highly functional, while a rating of 4 is functional. Similarly, a rating of 3 suggests moderately functional, a rating of 2 indicates less functional, and a rating of 1 indicates not functional.

Table 7. Rating Scale used in Evaluating the Reliable of the System.

Rating	Interpretation
5	Highly Reliable
4	Reliable
3	Moderately Reliable
2	Less Reliable
1	Not Reliable

Table 7 presents the rating scale used to evaluate the *reliability* of the system. A rating of 5 indicates that the system is *highly reliable*, showing consistent performance without errors or failures. A rating of 4 means the system is *reliable*, functioning well under normal operating conditions. A rating of 3 suggests the system is *moderately reliable*, performing adequately but with minor issues. A rating of 2 reflects that the system is *less reliable*, experiencing occasional inconsistencies or errors, while a rating of 1 means the system is *not reliable* and frequently fails to perform its intended functions.

Table 8. Rating Scale used in Evaluating the User Convenience of the System.

Rating	Interpretation
5	Highly Convenient
4	Convenient
3	Moderately Convenient
2	Less Convenient
1	Not Convenient

Table 8 presents the rating scale used to evaluate the *user convenience* of the system. A rating of 5 indicates that the system is *highly convenient*, meaning it is very easy to use, accessible, and user-friendly. A rating of 4 signifies that the system is *convenient*, functioning smoothly with minimal effort required from the user. A rating of 3 suggests that the system is *moderately convenient*, offering an acceptable level of ease but with some aspects that could be improved. A rating of 2 denotes that the system is *less convenient*, requiring more effort or time to

operate efficiently, while a rating of 1 indicates that the system is *not convenient*, difficult to use, and may cause frustration or confusion among users.

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<https://repository.umindanao.edu.ph/handle/20.500.14045/1247>

APPENDICES

PLAN OF COURSE WORK

Name: **RENA ROSE L. BACO**Course: **BSCpE**

Major Course:

Course No.	Course Description	Unit	Remarks
CpE 311	Computer System Organization with Assembly Language	4	
CpE 211	Computer Fundamental and Programming	3	
CpE 212	Computer Hardware Fundamentals	1	
CpE 221	Data Structures and Algorithm Analysis	4	
CpE 321	Computer Engineering Drafting and Design	1	
CpE 322	Logic Circuit Switching Theory	4	
CpE 411	PC Troubleshooting & Repair	3	
CpE 412	Advance Logic Circuits	4	
CpE 413	Control System	4	
CpE 421	Operating System	4	
CpE 422	Microprocessors System	4	

Republic of the Philippines
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College of Engineering
Isulan Campus
Isulan, Sultan Kudarat

CpE 425	Elective 1	3	

Total Number of Units Required for the Period : **75 UNITS**

Total Numbers of Units Earned : **## Units**

Certified Correct:

NAME OF REGISTRAR

Campus Registrar

Date

Approved:

ROMMEL M. LAGUMEN

Campus Director

Date

APPLICATION FOR THESIS TITLE

**SMART PET FEEDING DEVICE
USING SMS**

Date

Remarks

Signature

COIN OPERATED VENDO PAPER

Remarks

Signature

**DRIED GRAIN CROPS BAGGING
MACHINE**

Remarks

Signature

We are planning to present our thesis title on August 22, 2025 at SKSU Isulan Campus.

Very respectfully yours,

NATHANIEL C. LAGNAODA

GEORGE ANDRIE D. GATAN
Students

Recommending Approval:

BEN ARNEL
Member

MEILAFLORE A. PACLIBAR, MEP-CoE
Member

JADE A. BUENAVIDES
Adviser

Endorsed:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

LENMAR T. CATAJAY, ME-CpE
College Dean

Date Signed

Date Signed

Approved:

ROMMEL M. LAGUMEN
Campus Director

Date Signed

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Isulan, Sultan Kudarat

NOMINATION OF GUIDANCE COMMITTEE

We, **NAME OF RESEARCHER**, students of BSCpE hereby nominate the following as adviser and members of our thesis guidance committee.

NAME OF ADVISER

Adviser

NAME OF MEMBER1

Member

NAME OF MEMBER2

Member

We, hereby certify our willingness to act as adviser / members of the guidance committee.

NAME OF ADVISER

Adviser

NAME OF MEMBER1

Member

NAME OF MEMBER2.

Member

Endorsed:

ELMER C. BUENAVIDES, DIT

Campus Research Coordinator

Date Signed

LENMAR T. CATAJAY, ME-CpE

College Dean

Date Signed

Approved:

ROMMEL M. LAGUMEN

Campus Director

Date Signed

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Isulan, Sultan Kudarat

APPLICATION FOR THESIS OUTLINE DEFENSE

Name: **NATHANIEL C. LAGNAODA**
GEORGE ANDRIE D. GATAN

Course/Major: **BSCpE**

We have the honor to apply for outline defense for our study entitled: “**SELF-SERVICE PRINTING KIOSK IN SKSU – ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH**”.

Time: _____
Date: _____
Venue: _____

BEN ARNEL
Member

MEILAFLORE A. PACLIBAR
Member

BEN ARNEL
Statistician

EDRALIN D. MESIAS
English Critic

JADE A. BUENAVIDES
Adviser

Endorsed:

Recommending Approval:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

LENMAR T. CATAJAY, ME-CpE
College Dean

Approved:

ROMMEL M. LAGUMEN
Campus Director

Remove this page if NOT used!

CHANGE OF ADVISER / GUIDANCE COMMITTEE MEMBER

Name: NAME OF RESEARCHER

Course: BS

Major: COURSE/MAJOR

Thesis Title: TITLE OF THE STUDY

APPROVED BY THE GUIDANCE COMMITTEE

NAME OF ADVISER

Adviser

Signature

Date

NAME OF MEMBER

Member

Signature

Date

NAME OF MEMBER

Member

Signature

Date

NAME OF STATISTICIAN

Statistician

Signature

Date

NAME OF ENGLISH CRITIC

English Critic

Signature

Date

Endorsed:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

Recommending Approval:

LENMAR T. CATAJAY, ME-CpE
College Dean

Approved:

ROMMEL M. LAGUMEN
Campus Director

APPROVAL OF THESIS OUTLINE

Name: NAME OF RESEARCHER
Thesis Title: [Click here to enter text.](#)

Course/Major: BS

APPROVED BY THE GUIDANCE COMMITTEE

NAME OF ADVISER
Adviser

Signature

Date

NAME OF MEMBER
Member

Signature

Date

NAME OF MEMBER
Member

Signature

Date

NAME OF STATISTICIAN
Statistician

Signature

Date

NAME OF ENGLISH CRITIC
English Critic

Signature

Date

Endorsed:

Recommending Approval:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

LENMAR T. CATAJAY, ME-CpE
College Dean

Approved:

ROMMEL M. LAGUMEN
Campus Director

CERTIFICATION OF STATISTICIAN

This is to certify that the thesis entitled INSERT TITLE OF THE STUDY conducted on _____, authored by INSERT NAME OF RESEARCHER was evaluated/checked by the undersigned as to the statistical analysis and interpretation.

Issued on this _____ day of _____.

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Statistician

Noted:

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APPLICATION FOR THESIS FINAL DEFENSE EXAMINATION

Name: **NAME OF RESEARCHER**

Course/Major: **BS**

Major: Click here to enter text.

Thesis Title: Click here to enter text.

Please write ✕ whether: () First () Second () Third
 Date: _____ Time: _____ Venue: _____

Guidance Committee

Name	Signature	Date
<u>NAME OF ADVISER</u> Adviser	_____	_____
<u>NAME OF MEMBER</u> Member	_____	_____
<u>NAME OF MEMBER</u> Member	_____	_____
<u>NAME OF STATISTICIAN</u> Statistician	_____	_____
<u>NAME OF ENGLISH CRITIC</u> English Critic	_____	_____

Endorsed:

Recommending Approval:

ELMER C. BUENAVIDES, DIT
Campus Research Coordinator

LENMAR T. CATAJAY, ME-CpE
College Dean

Approved:

ROMMEL M. LAGUMEN
Campus Director

Date

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Report on the Result of Final Defense

(Action taken by the Guidance Committee. Please indicate whether Passed or Failed)

Signature	Date	Remarks

Approved:

ROMMEL M. LAGUMEN
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APPLICATION FOR THESIS FINAL PRINTING AND BINDING

This is to certify that the thesis entitled **INSERT TITLE OF THE STUDY**
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English Critic

Date Signed**NAME OF STATISTICIAN**

Statistician

Date Signed**NAME OF MEMBER**

Member

Date Signed**NAME OF MEMBER**

Member

Date Signed**NAME OF ADVISER.**

Adviser

Date Signed

Recommending Approval:

ELMER C. BUENAVIDES, DIT

Campus Research Coordinator

Date Signed**LENMAR T. CATAJAY, ME-CpE**

College Dean

Date Signed

Approved:

ROMMEL M. LAGUMEN

Campus Director

Date Signed

SAMPLE QUESTIONNAIRE

RESEARCH INSTRUMENT

SELF-SERVICE PRINTING KIOSK IN SKSU(SSPK) – ISULAN LIBRARY WITH CASHLESS PAYMENT INTEGRATION VIA GCASH

Name (Optional): _____

Instruction: Rate the self-service printing kiosk (SSPK) from 1 to 5 according to rating scale. Check the table which correspond to your evaluation.

Rating scale to be used in evaluating the SSPK system

Rating	Interpretation
5	Strongly Agree
4	Agree
3	Moderately Agree
2	Disagree
1	Strongly Disagree

I. Functionality					
Questions	Ratings				
The SSPK system is functional in terms of:	5	4	3	2	1
1. The SSPK System boots without error.					
2. The system interface is responsive and user-friendly					
3. The file uploading operates smoothly.					
4. The GCash payment process functions properly.					
5. The system completes printing accurately.					

II. Reliability					
Questions	Ratings				
The SSPK system is reliability in terms of:	5	4	3	2	1
1. The system consistently performs printing and payment transactions without failure.					
2. The kiosk responds promptly to user actions without delays or unresponsiveness					
3. The kiosk remains functional and accessible throughout library hours.					
4. Payments made through GCash are accurately verified and recorded.					
5. The kiosk provides accurate output according to the user's selected file.					

III. User Convenience					
Questions	Ratings				
The SSPK system is user convenience in terms of:	5	4	3	2	1
1. The kiosk is easy to navigate and understand even for first-time users.					
2. The cashless payment option through GCash makes transactions faster and easier.					
3. The kiosk is accessible and available for use when needed.					
4. The kiosk allows me to complete printing tasks quickly and efficiently.					
5. Overall, the kiosk provides a convenient and satisfying user experience.					

PICTORIALS AND SCREENSHOTS

SOURCE CODE

DVD